Loïc Guezou

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website | Google Scholar | ORCID | LinkedIn

Current position

Postdoctoral researcher, Institute of Material Systems Modeling, Helmholtz-Zentrum Hereon, Hamburg area, Germany
Since: October 2026

Research interests

Scientific Machine Learning; Model Order Reduction; Constitutive Modelling; Biomechanics; Non-linear Materials

Education

Ph.D. in Computational Mechanics
Université Paris-Saclay 2022 – 2026

Thesis: *Development of Hyperelastic Reduced-Order Models For Biomimetic Material Applications Using Proper Generalised Decomposition*
Defended: 16 July 2026. **Supervisors**: B. Tie, A. Barbarulo
Jury: A. Bel-Brunon (president), P. Díez, U. Nackenhorst (reviewers), A. Daby-Seesaram

M.Sc. in Computational Mechanics
Université Paris-Saclay 2021 – 2022

M.Sc. in Mechanical Engineering
ISAE-Supméca 2019 – 2022

CPGE (preparatory cycle to the top Engineering schools entrance exams)
Lycée Condorcet, Paris 2017 – 2019

Experience

Postdoctoral researcher Hamburg area, Germany
Institute of Material Systems Modeling, Helmholtz-Zentrum Hereon October 2026 – present

Ph.D. student in Computational Mechanics Gif-sur-Yvette, France
Laboratoire de Mécanique Paris-Saclay October 2022 – July 2026

Skills developed: Model order reduction (PGD), Hyperelasticity theory, Python, Finite Element Method, Numerical methods, Biomimetic Engineering, Scientific writing

Teaching Gif-sur-Yvette, France
CentraleSupélec 2023 – 2025 (24h)

Teaching assistant for Computational Structural Dynamics. Master 2 students. 2 years in a row.

Teaching Gif-sur-Yvette, France
CentraleSupélec 2022 – 2024 (40h)

Project supervision on Mechanical Engineering for 3D printing. Master 1 Engineering students. 2 years in a row.

Research intern in Computational Mechanics Gif-sur-Yvette, France
Laboratoire de Mécanique Paris-Saclay April 2022 – September 2022

Master research internship. Developing a PGD dual formulation (pressure-velocity) for acoustic dynamic computations.
Supervision: B. Tie, A. Barbarulo

Engineer assistant intern in Research & Development Morestel, France
NUVIA Protection September 2020 – January 2021

Quantification of the head loss in fire protection devices for the nuclear sector.

Student intern Paris, France
Laboratoire de Physique de l'Ecole Normale Supérieure January 2020

Publications

1. Loïc Guezou, Andrea Barbarulo, Bing Tie. A new PGD parametric modelling strategy for hyperelastic materials. *In preparation.*
2. Loïc Guezou. Development of Hyperelastic Reduced-Order Models For Biomimetic Material Applications Using Proper Generalised Decomposition. *PhD thesis, Université Paris-Saclay*, 2026; theses.fr.

3. Loïc Guezou, Andrea Barbarulo, Bing Tie. Development of high-fidelity numerical models for 3D printing of patient-specific anatomy. *ECCOMAS Congress 2024 - The 9th European Congress on Computational Methods in Applied Sciences and Engineering*, Lisbon, Portugal; Abstract.
4. Loïc Guezou, Andrea Barbarulo, Bing Tie. Development of high-fidelity numerical models for 3D printing of patient-specific anatomy. *16ème Colloque National en Calcul de Structures (CSMA 2024)*, Hyères, France; Abstract.

Talks

1. Loïc Guezou. PhD defence. *Université Paris-Saclay*, 2026 July 16; Gif-sur-Yvette, France.
2. Loïc Guezou, Andrea Barbarulo, Bing Tie. Poster presentation. *7th International Workshop on Model Order Reduction Techniques*, 2025 November 26; Zaragoza, Spain; Poster.
3. Loïc Guezou, Andrea Barbarulo, Bing Tie. Presentation. *9th European Congress on Computational Methods in Applied Sciences and Engineering*, 2024 June 07; Lisbon, Portugal; Talk.
4. Loïc Guezou, Andrea Barbarulo, Bing Tie. Poster presentation. *16th Congress of the Computational Structural Mechanics Association*, 2024 May 14; Giens, France; Poster.

Skills

Languages: French (native) | English (fluent, TOEIC 980/990) | German (A1, learning) | Spanish (intermediate) | Italian (intermediate)
Programming tools: Python | C++ | MatLab
Methods: Finite Element Method | Model Order Reduction | Neural Networks
Scientific Software: CATIA | SolidWorks
OS: macOS | Linux | Windows
Other software: LaTeX | Typst | Inkscape | Microsoft Office

Service

ED SMEMaG, Elected representative of the PhD students in the doctoral school council 2022–2024